

SMART PARKING SYSTEM — PROJECT SYNOPSIS

Phase 2 Frontend Development: Responsive UI, Forms & Client-Side Validation

GitHub Repository: https://github.com/Chaitya44/Smart_Parking_System

1. EXECUTIVE OVERVIEW & REPOSITORY LINK

The Smart Parking System is a modern, responsive web application built to eliminate parking congestion and streamline facility operations. Key capabilities include real-time slot occupancy visualization, dynamic bay allocation, EV charging station indicators, automated entry ticketing, and instant checkout with digital revenue tracking.

Submission Metadata:

- Repository Link: https://github.com/Chaitya44/Smart_Parking_System
- Phase: Phase 2 (UI Forms & Client Validation)
- Branch: main
- Strict Zero Comments: Verified

2. CORE DELIVERABLES BUILT (PHASE 2)

A. Responsive Home / Landing Page (basic/html/index.html & basic/css/home.css)

Features a modern hero section with live capacity preview widget, 4-pillar features grid (Real-Time Visualizer, EV Charging Stations, Multi-Category Zones, Automated Billing), a 3-step workflow guide ("How It Works"), CTA action banners, and sticky navigation.

B. Management Dashboard (basic/html/dashboard.html & basic/css/dashboard.css)

Full operations center featuring 5 KPI summary cards (Total Bays, Available, Occupied, Reserved, Today's Revenue). Includes a spacious real-time slot grid with grey vehicle category icons, EV charging badges, quick filter toolbars, and an audit table.

C. Client-Side Live Form Validation (basic/js/login.js, register.js, dashboard.js)

Active validation on both blur and live input events across all forms. Includes real-time password strength meter (length, uppercase, lowercase, numbers, symbols), Indian mobile regex (10 digits starting 6-9), vehicle license plate pattern, and duplicate slot ID prevention.

D. Modular Directory Layout & PHP Readiness (basic/css/, html/, js/, php/)

Cleanly reorganized into dedicated modular folders. Includes prepared PHP backend templates (db.php, login.php, register.php) ready for Phase 3 database integration.

3. SYSTEM ARCHITECTURE & FILE BREAKDOWN

Component	File Path	Key Responsibilities
Home Landing Page	basic/html/index.html	Value proposition, live capacity widget, features grid, navigation
Operations Dashboard	basic/html/dashboard.html	5 KPI cards, real-time slot visualizer grid, modals, activity table
User Authentication	basic/html/login.html , register.html	Login/registration forms, password visibility toggle, strength meter
Client Logic Layer	basic/js/dashboard.js	ParkingAPI service layer, slot allocation, ticket generation, checkout
Styling & Aesthetics	basic/css/ (home, dashboard, login)	Soft rounded corners (16-20px), responsive grids, clean pill badges
Backend Templates	basic/php/ (db, login, register.php)	Prepared database connection and authentication endpoint scripts

SMART PARKING SYSTEM — TECHNICAL SPECIFICATIONS

Feature Highlights, Validation Criteria & Next Phase Roadmap

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4. IN-DEPTH FEATURE HIGHLIGHTS

• Real-Time Interactive Slot Visualizer Grid

Dynamic slot cards with minimum 260px width eliminating awkward text squishing. Each card displays Slot ID, Zone, Category icon (Car, Bike, Scooter, EV), EV station indicator pill (EV), and status badge (Available, Occupied, Reserved). Occupied spots display the active license plate, driver name, and arrival time.

• Automated Entry Ticketing & Digital Checkout

One-click modal workflow allowing operators to park incoming vehicles by selecting an available bay, validating license plate format, and applying category rates (Rs.40/hr car, Rs.20/hr bike, Rs.60/hr EV). The checkout modal calculates elapsed duration, computes fees, and supports Cash, UPI QR, and Fastag.

• "+ Add Parking Slot" Capacity Management

Enables parking lot administrators to expand parking inventory in real time. Validates custom Slot IDs against existing slots to prevent duplicates, assigns parking zones, sets vehicle categories, and configures EV station availability.

• Instant Multi-Criteria Search & Filter Engine

Instant client-side filtering by Search Query (Slot ID, License Plate, Driver Name), Zone (Zone A, B, C, D), Status (Available, Occupied, Reserved), and EV Station capability (EV Charging Only vs Standard).

5. CLIENT-SIDE FORM VALIDATION MATRIX

Field	Validation Constraint	RegEx / Evaluation Method
Full Name / Driver Name	3 to 60 letters and spaces only	<code>^[A-Za-z\s]{3,60}\$</code>
Email Address	RFC-compliant email pattern	<code>^[^\s@]+@[^\s@]+\.[^\s@]+\$</code>
Mobile Phone Number	10-digit Indian mobile starting with 6, 7, 8 or 9	<code>^[6-9]\d{9}\$</code>
Password Strength	Min 8 chars, uppercase, lowercase, numeric, special	Live 3-Tier Meter
Vehicle Registration	6-15 alphanumeric characters with spaces or hyphens	<code>^[A-Za-z0-9\s-]{6,15}\$</code>
Slot ID Format	2-10 alphanumeric characters with unique ID check	<code>^[A-Za-z0-9-]{2,10}\$</code>

6. CODE QUALITY & STANDARDS ADHERENCE

- Strict Zero-Comments Rule: Verified through recursive automated scanning. Exactly 0 comments exist across all HTML, CSS, JS, and PHP files.
- Separation of Concerns: Clean directory modularity with separate `css/`, `html/`, `js/`, and `php/` folders.
- Database-Ready Architecture: ParkingAPI service layer returns Promises, allowing seamless future transition to PHP/MySQL.
- Cross-Device Responsiveness: Fully tested across mobile smartphones (320px+), tablets, and desktop resolutions.

7. FUTURE ROADMAP (PHASE 3 & 4)

1. Database Provisioning: Create normalized MySQL tables for users, parking_slots, and activity_logs.
2. PHP API Endpoints: Replace local mock data with asynchronous fetch/AJAX calls to backend PHP handlers.
3. Automated Payment Gateway: Integrate digital payments (Stripe / Razorpay / UPI QR generation).